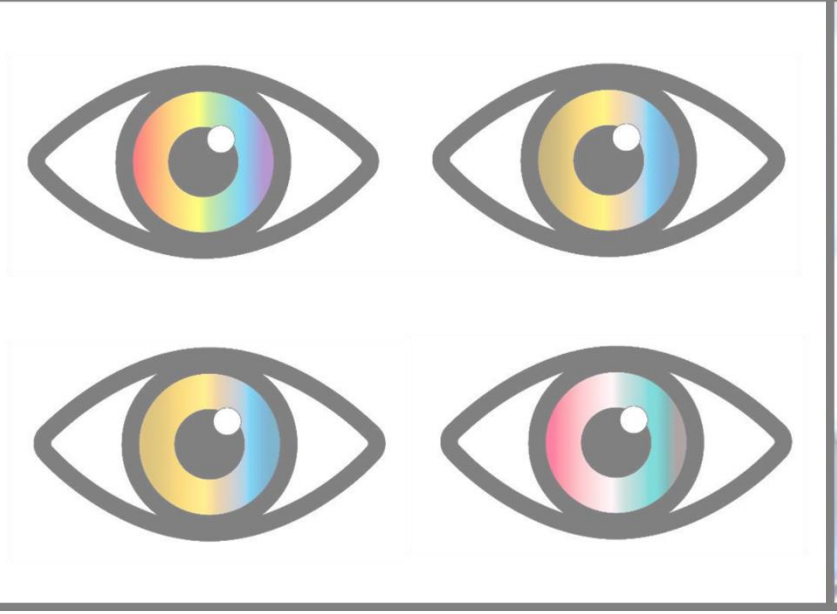

Colour Vision Deficiency in Chemistry: Challenges and Practical Solutions

Nicholas J. Roberts, Dr. Jennifer L. MacDonald
Dalhousie University, Department of Chemistry

Canadian Chemistry Conference and Exhibition
June 8th, 2023

The Colours Project

The Team

Nick Roberts

Katlyn Near

Rory Chisholm

Jenn MacDonald

Mark Wall

Funding

- LearnSci/CLEAR21 Grant
- Dalhousie University
  - Department of Chemistry
  - Centre for Learning and Teaching
  - Office of Accessibility and Accommodation

Colour Vision Deficiency (CVD)

- CVD affects ~9% of the world's population
- Difficulty or inability to distinguish certain colours
- Colour vision is dependent on **cones**

CVD Type	Lack of Pigment or Shift in Receptor	Commonly Confused Colours
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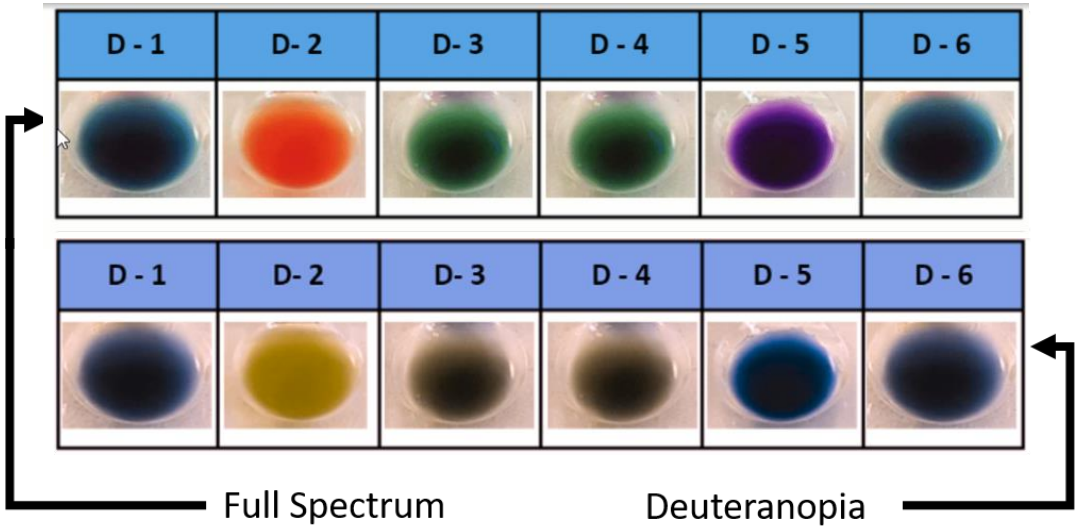
Deuteranopia	Green (Medium λ)	Red, Green, Blue, Purple
Tritanopia	Blue (Short λ)	Blue, Green, Yellow

(1) Colour Blind Awareness, (2) McIntyre, (3) Chromatic Vision Simulator

CVD in the First-Year Chemistry Laboratory (1)

- Potentially ~105 students in our class and ~3 teaching staff
- 10 out of 16 First-Year Chemistry experiments have some aspect of colour-based data/observations

Universal Indicator



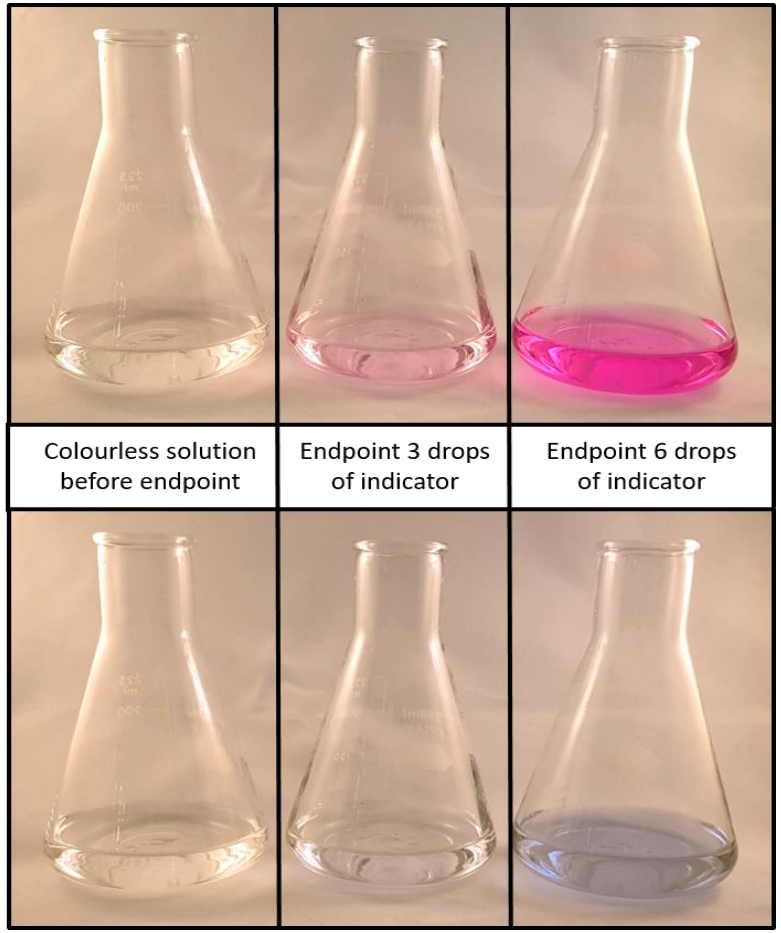
Column Chromatography

Full Spectrum Deuteranopia

(5) Coblis Colour Blindness Simulator

CVD in the First-Year Chemistry Laboratory (2)

Full Spectrum



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"**I feel like I am cheating** if I ask you to confirm that I have assigned my colours correctly."

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- Don't eliminate colour entirely, instead...
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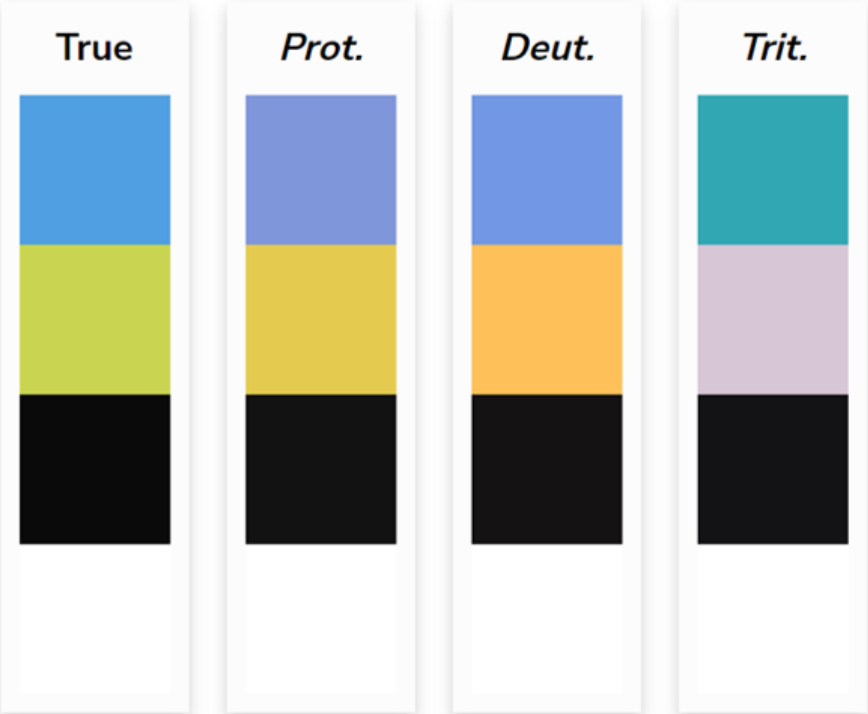

(4) Coloring for Colorblindness

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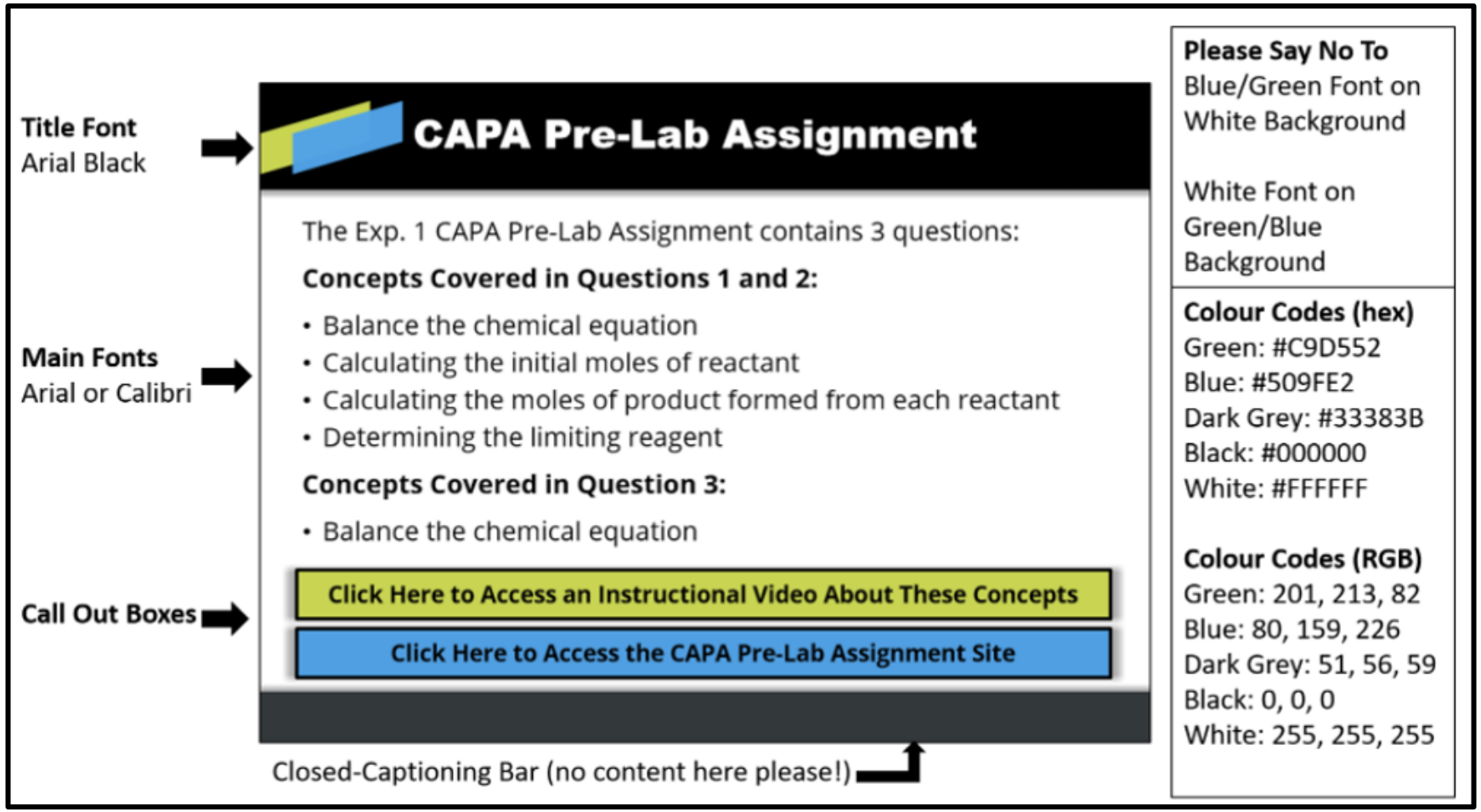
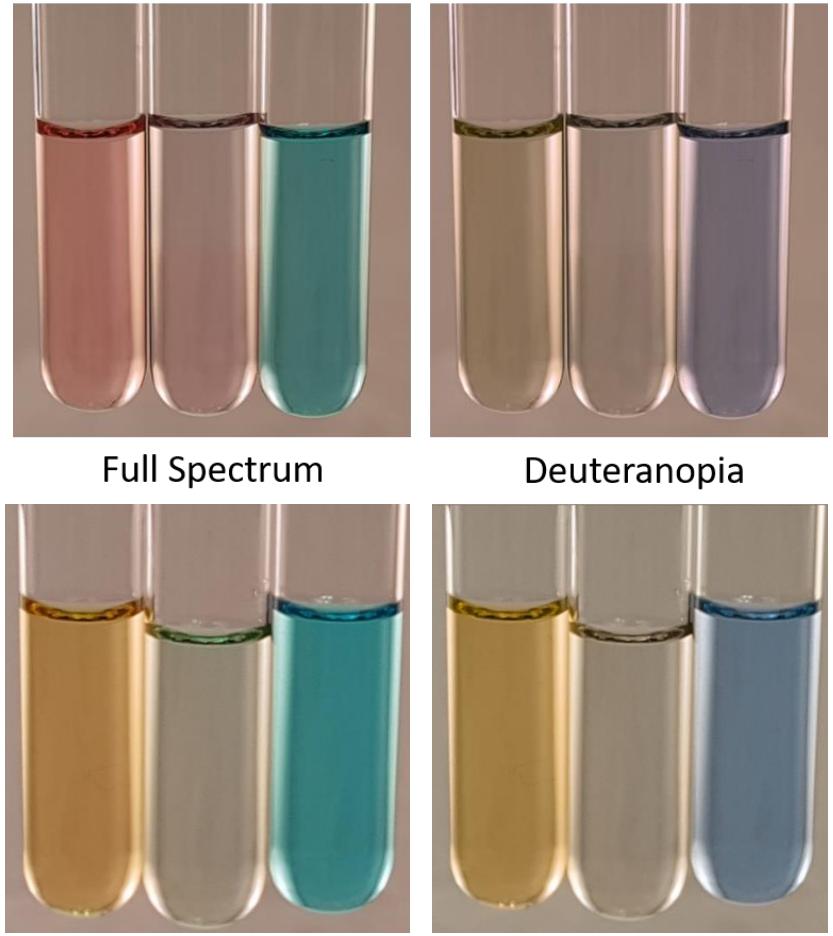
- Share with your team if you are co-developing content



(4) Coloring for Colorblindness, (5) Coblis Colour Blindness Simulator, (6) WebAIM

Supporting Students and Educators (1)

Column Chromatography



- **Student Facing:** Same experiment, option for CVD Friendly unknown

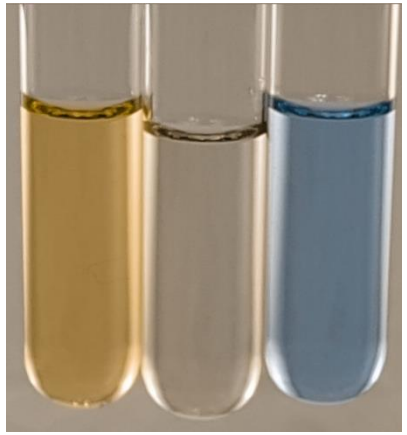
- **Lab Manual Text:** If you or a teammate suffers from colour vision deficiency (CVD), feel free to ask the teaching team for an unknown with a dye colour mixture that supports red/green CVD or blue/yellow CVD.

Supporting Students and Educators (2)

Column Chromatography

Full Spectrum

Deuteranopia



- **Teaching Team Facing:** Custom pocket-sized note card linking unknowns to colours present (red, green, blue, yellow)

		Unknown Numbers		
Dye 1	Dye 2	Main	CVD Support	CVD Type
Red	Blue	2-digit unknown number	3-digit unknown number	Blue/Yellow CVD (Tritanopic)
Red	Green	2-digit unknown number		
Yellow	Blue	2-digit unknown number	3-digit unknown number	Red/Green CVD (Protanopic/ Deuteranopic)
Yellow	Green	2-digit unknown number		

Supporting Students and Educators (3)

Teaching Team Training Modules

Helpful Tips – Part B (3)

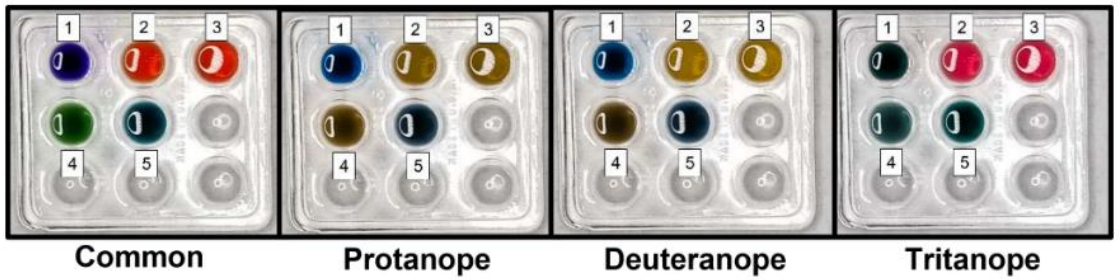
- **Phenolphthalein** typically causes no problems
- **Metacresol purple** may present a brown colour for neutral solutions (this is OK)
- **Universal indicator** may require additional indicator to resolve weak base (blue) and strong base (purple) unknowns

Universal Indicator
Hover over the pH paper to confirm colour identification

Helpful Tips – Part B (4)

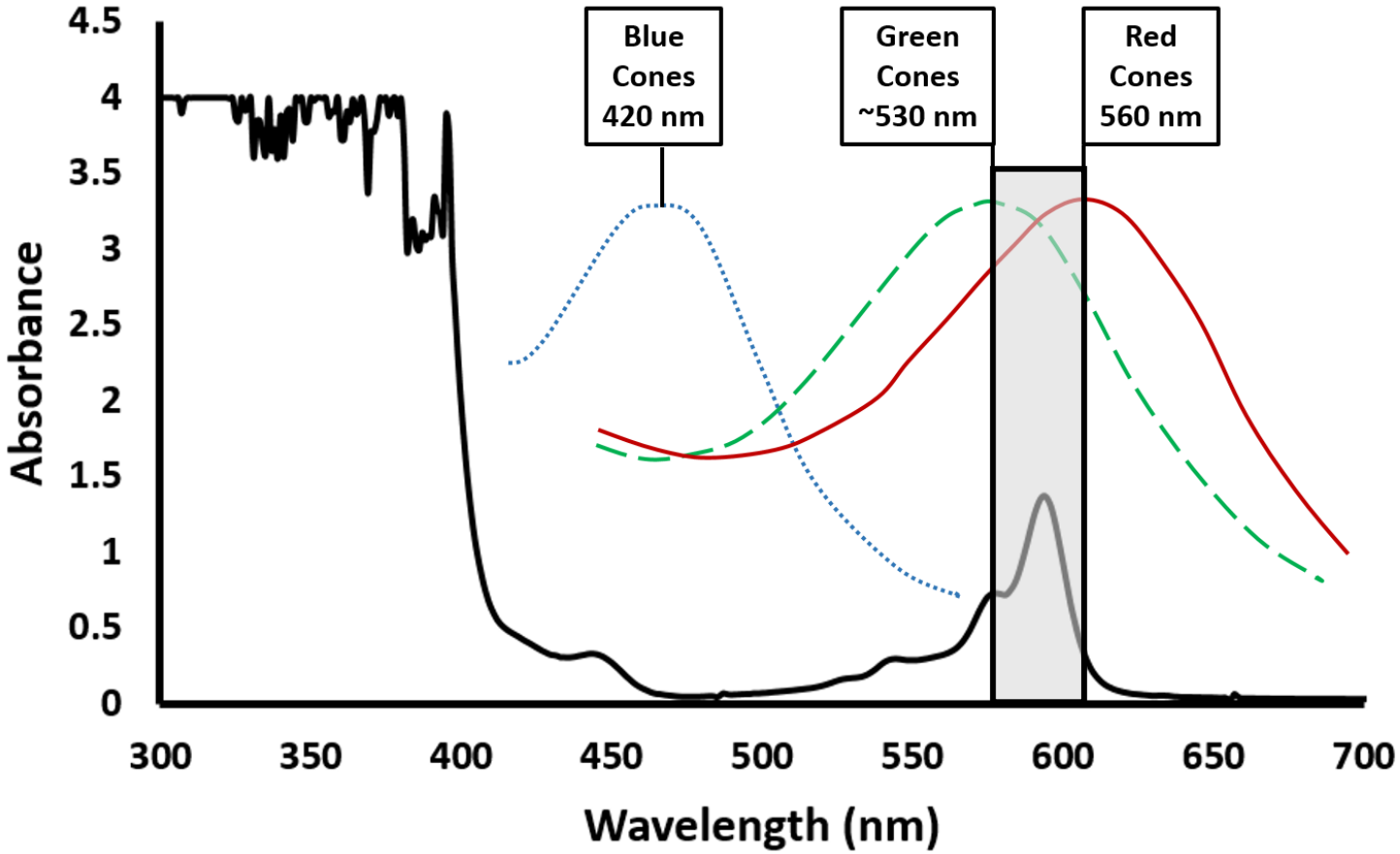
- Lab is reliant on full colour vision

- Colour Vision Deficient (CVD) students are welcome to ask for assistance



Commercially Available CVD Filter – EnChroma

- Deuteranopia: missing green cones
- EnChroma Glasses “block” wavelengths



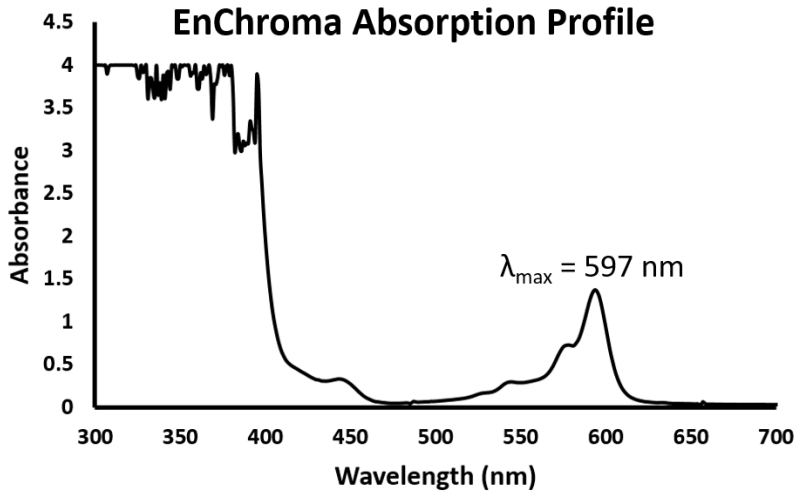
(2) McIntyre, (5) EnChroma, (6) We Are Colour Blind

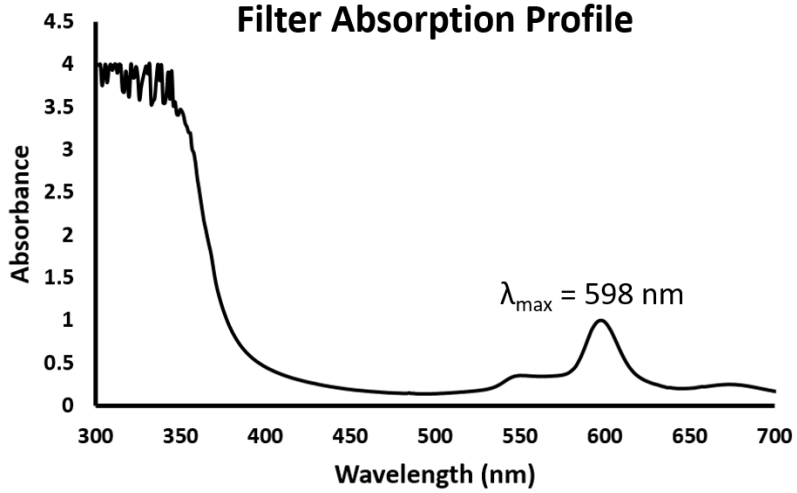
Preparing in-house CVD filter

- **Dye:**
  - ~10 mg ABS 594 (Luxottica)

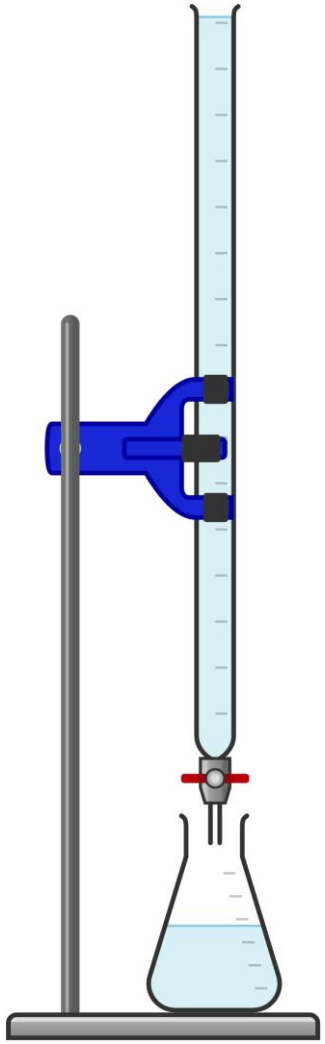
- **Solvent:**
  - 4 mL acrylic polyurethane

- **Casting:**
  - Drop Coat solution on Clean/Dry Glass Lens
  - Cure for 24 Hrs





How do we Quantitate Personal Colour Experience?

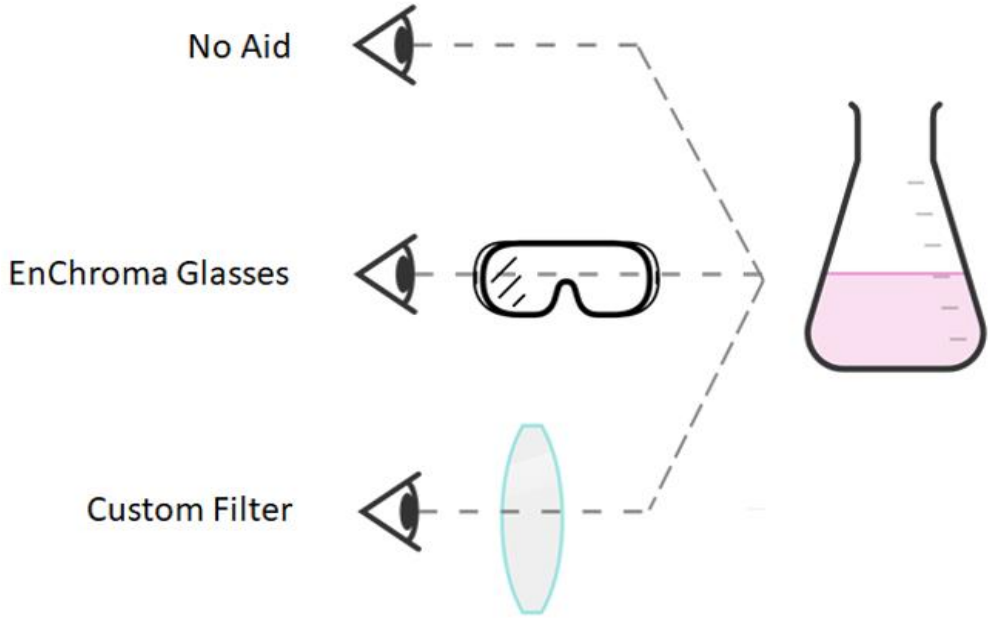


Potassium hydrogen phthalate (KHP) was titrated against standardized Sodium Hydroxide (NaOH)

~0.1 M NaOH
(standardized with pH probe)

~0.5g KHP
3 drops phenolphthalein (0.1%)
25 mL distilled water

- Two groups of individuals:
  - CVD: n = 2
  - Control: n = 2
- Order of titrating was randomized
- Mass of KHP unknown during experiment



Results

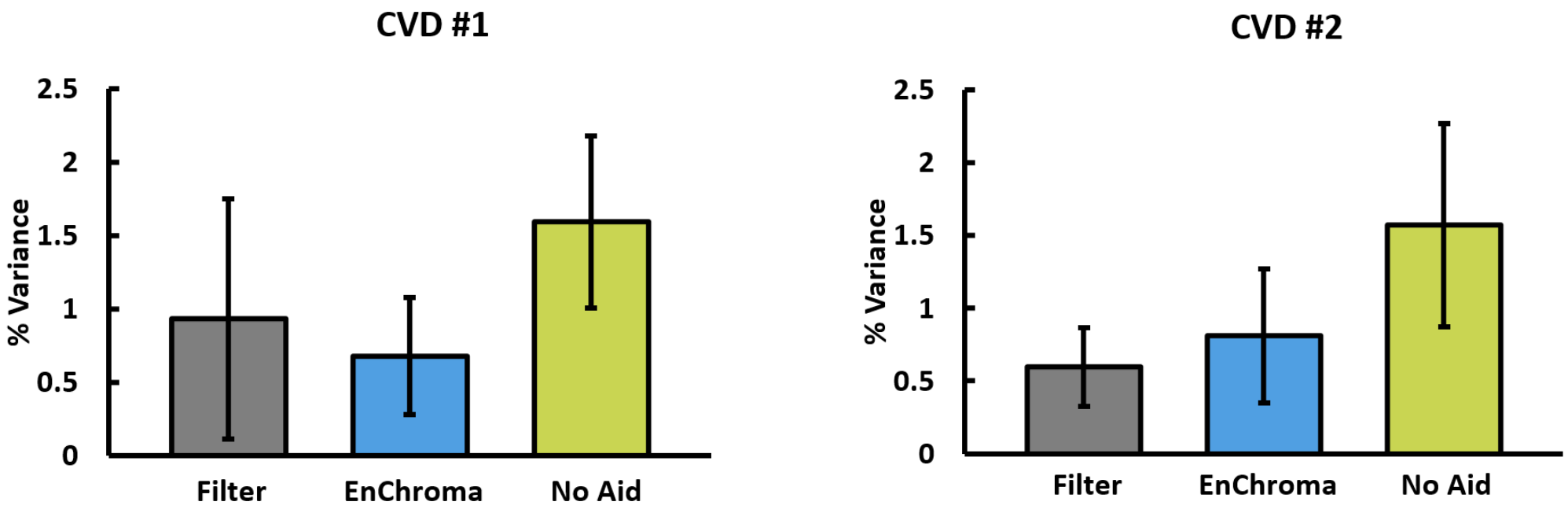


Figure 3: Comparison of % Variance in Moles of KHP From Actual Determined With Filter, EnChroma Glasses, and No Aid for CVD Individuals

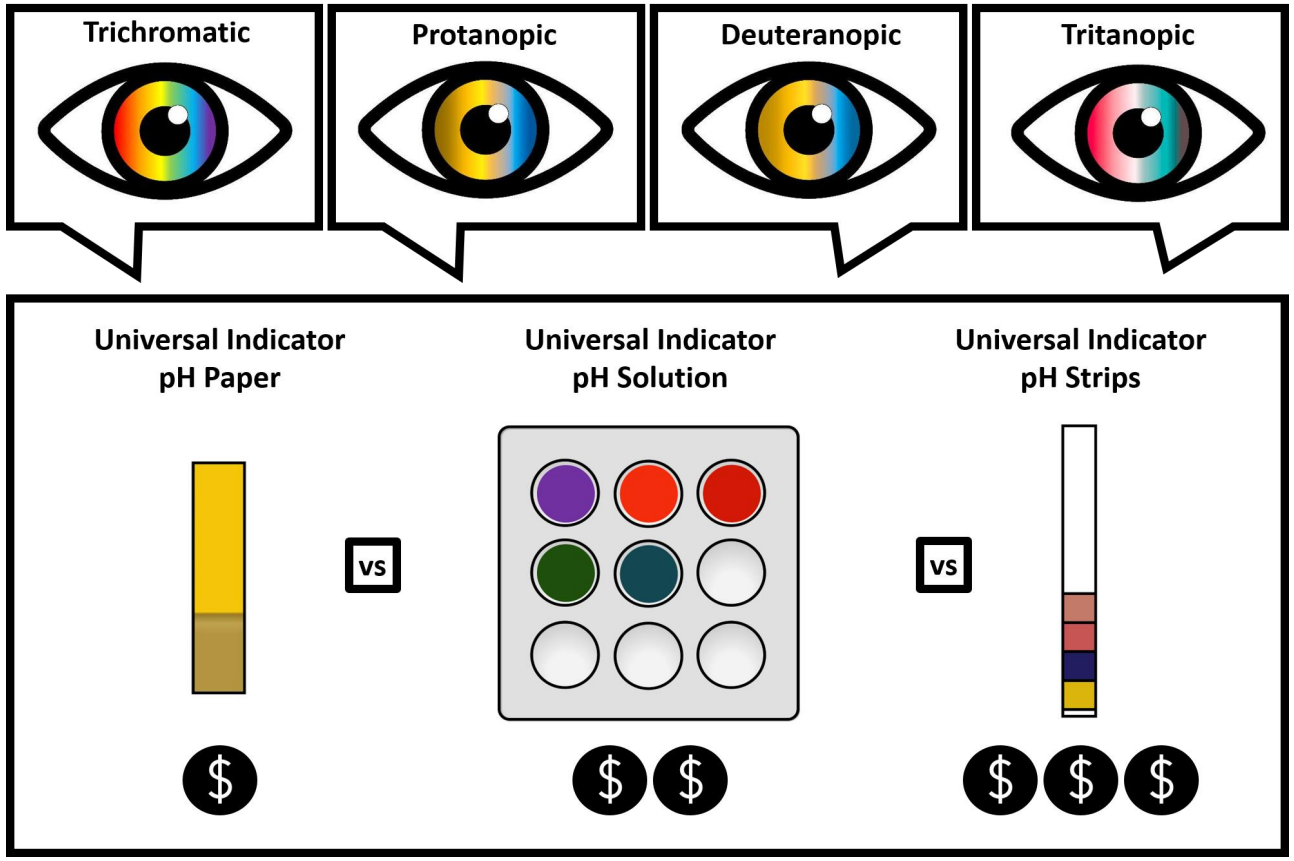
- CVD individuals observed similar visual benefits using the filter as compared to EnChroma glasses
- CVD individuals reported seeing the short-lived splotches of pink solution that signal the endpoint is near with the filter and EnChroma glasses

Students as Partners: Other Works

How Effective Are Indicators for Individuals with Colour Vision Deficiency?

N. J. Roberts, T. Hynes, D. Stacey, J. L. MacDonald (*ChemRxiv*, **2023**)

DOI: 10.26434/chemrxiv-2023-6g77r



Students as Partners: Other Works

Coming in 2024…an Open Educational Resource!

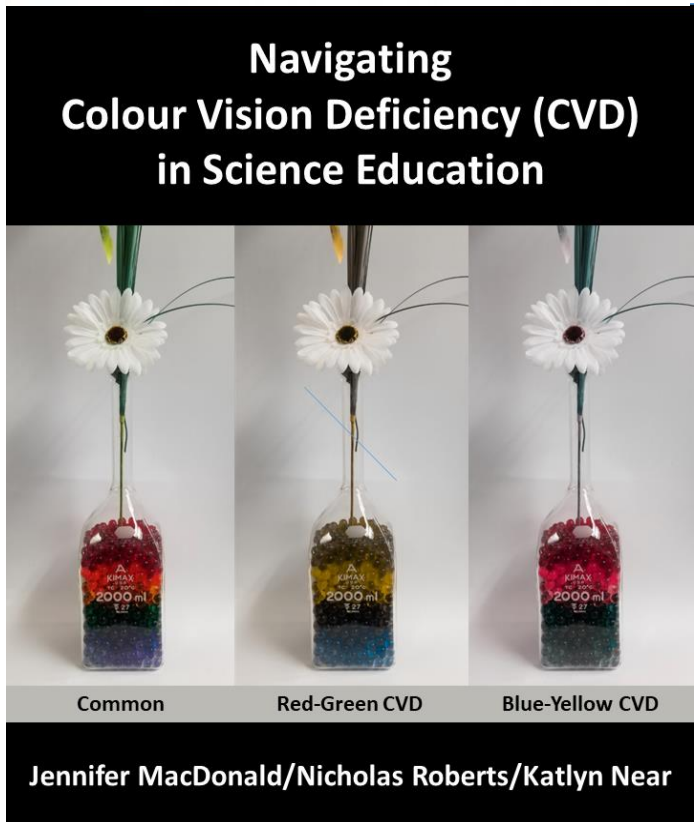
PB PRESSBOOKS

Navigating Colour Vision Deficiency (CVD) in Science Education

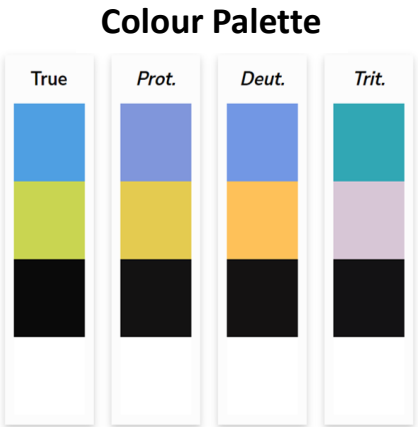
J. L. MacDonald, N. J. Roberts, K. G. Near

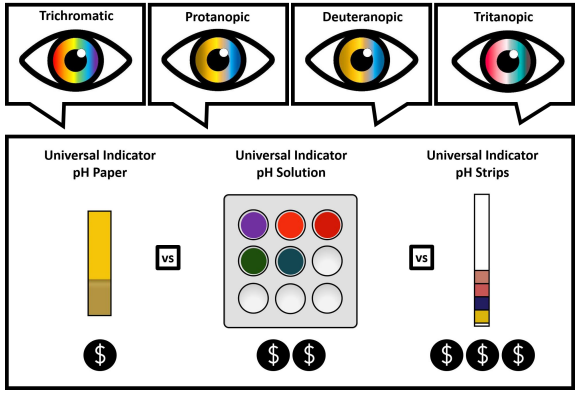
Add yourself to the mailing list to be notified when the OER is released.

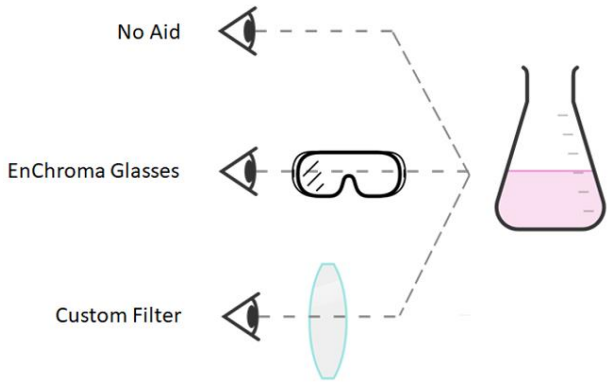
tinyurl.com/ColoursOER

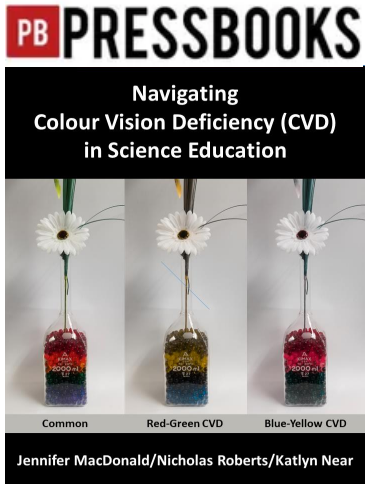


Creating Community









Awareness, Access, Inclusion and Belonging

Exclusionary Language	Inclusionary Language
"I feel like I am cheating..."	"Our team would like to opt for the colour-blind friendly unknown please."

References & Contact

References

1. Colour Blind Awareness https://www.colourblindawareness.org/ (accessed Feb 11, 2023).
2. McIntyre, D. Colour Blindness: Causes and Effects; Dalton Publishing: Chester, 2002.
3. Chromatic Vision Simulator App: https://asada.website/cvsimulator/e/ (accessed Feb 11, 2023).
4. Coloring for Colorblindness: https://davidmathlogic.com/colorblind/ (accessed Feb 11, 2023).
5. Coblis Colour Blindness Simulator Coblis — Color Blindness Simulator – Colblindor (color-blindness.com) (accessed Feb 11, 2023).
6. WebAIM (Web Accessibility in Mind) Contrast Checker: https://webaim.org/resources/contrastchecker/ (accessed Feb 11, 2023).
7. EnChroma Eyewear for Color Blindness https://enchroma.com/ (accessed Feb 11, 2023).

Feel free to contact us!

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Email: nicholasroberts@dal.ca

Website: njeffrob.github.io

Jenn

Email: jennifer.l.macdonald@dal.ca

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QR?

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LearnSci

#CLEAR22

DALHOUSIE UNIVERSITY

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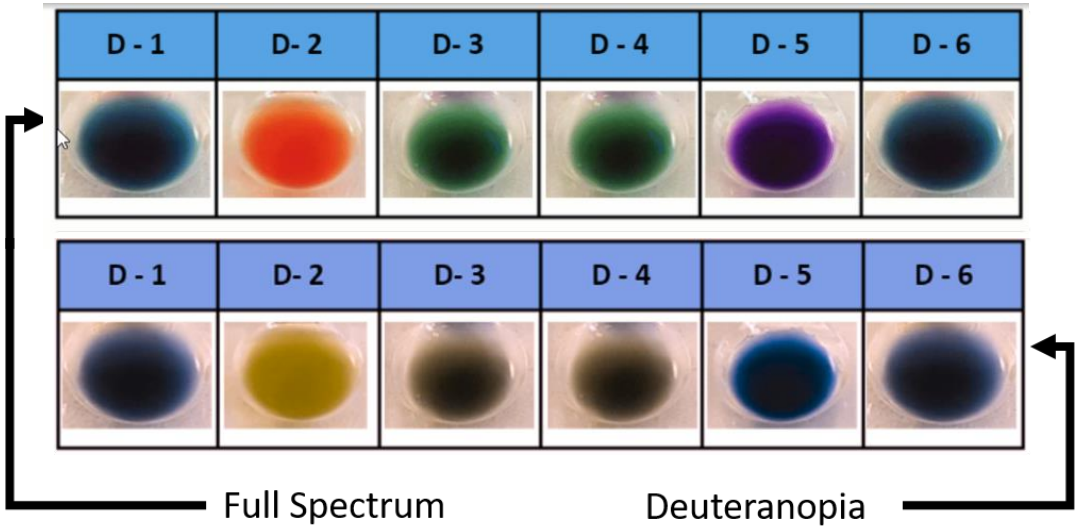

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Full Spectrum Deuteranopia

(5) Coblis Colour Blindness Simulator

CVD in the First-Year Chemistry Laboratory (2)

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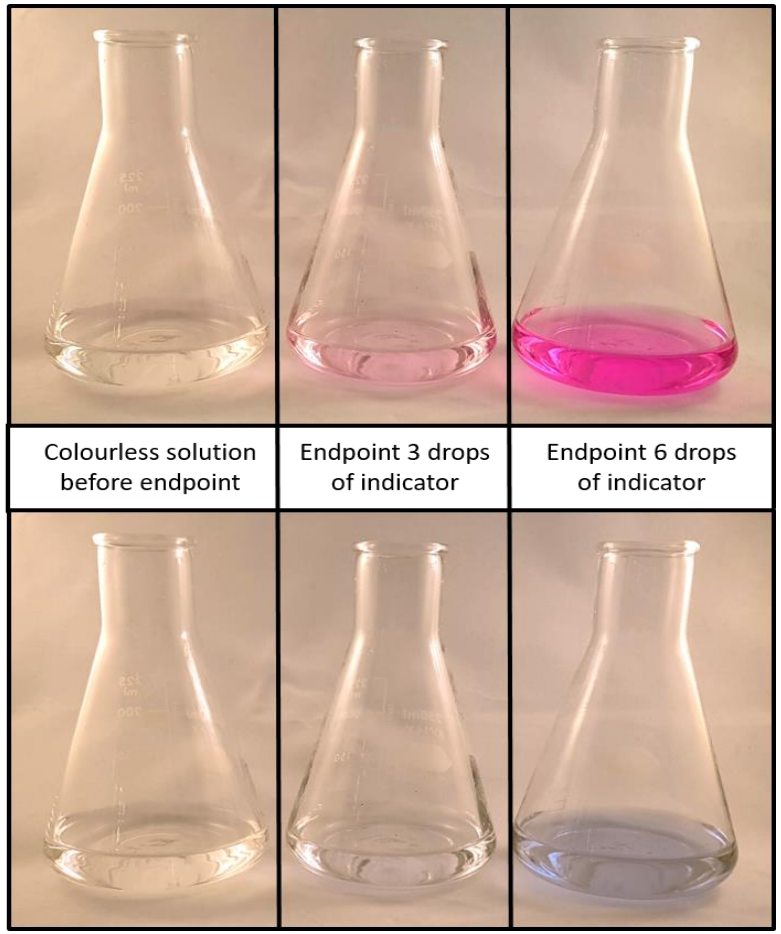

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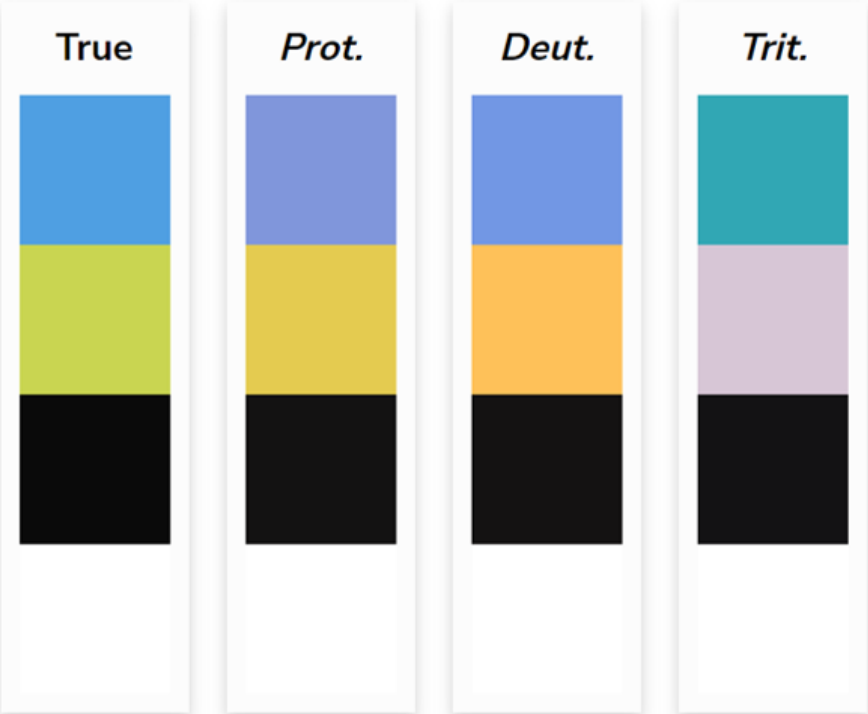

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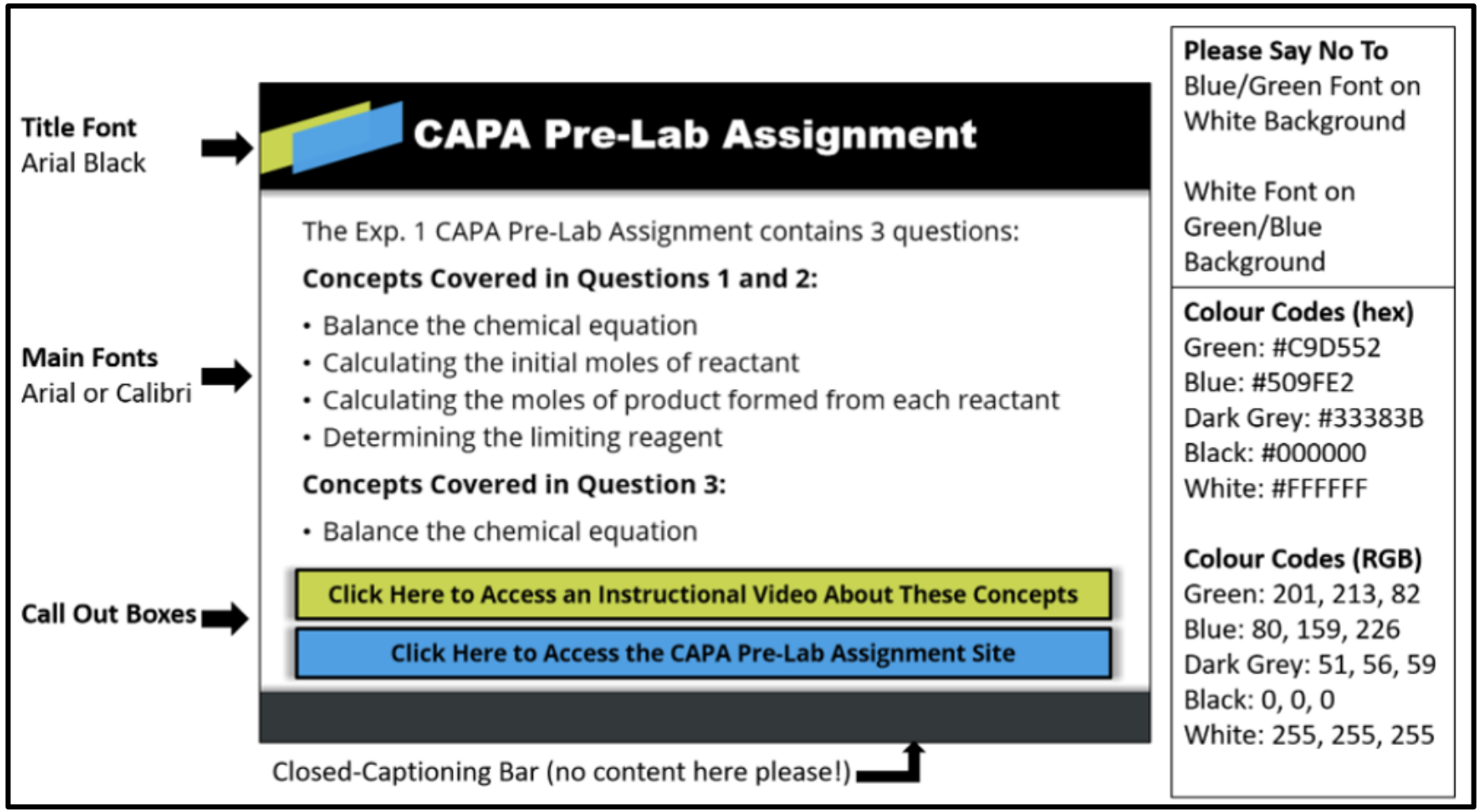

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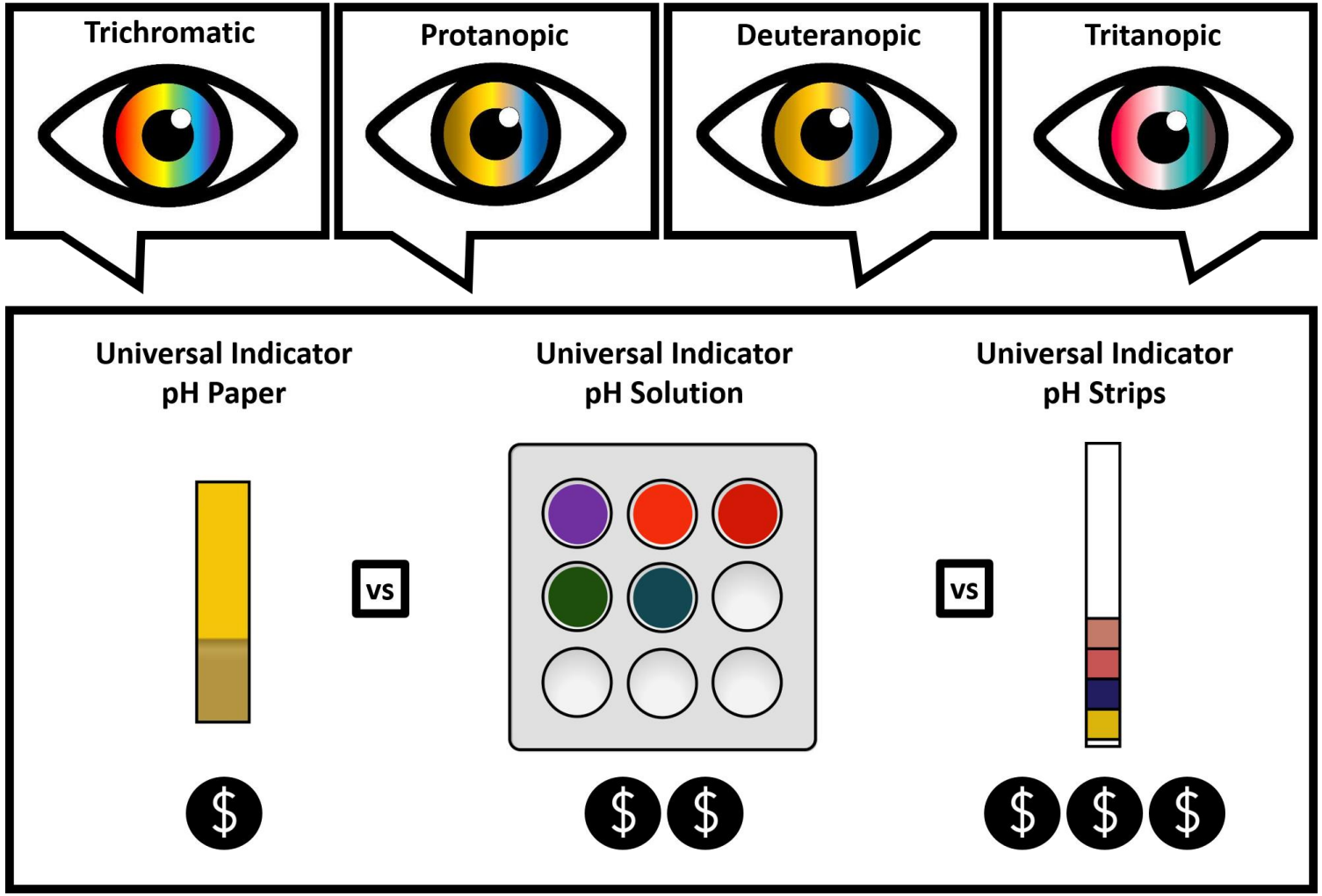
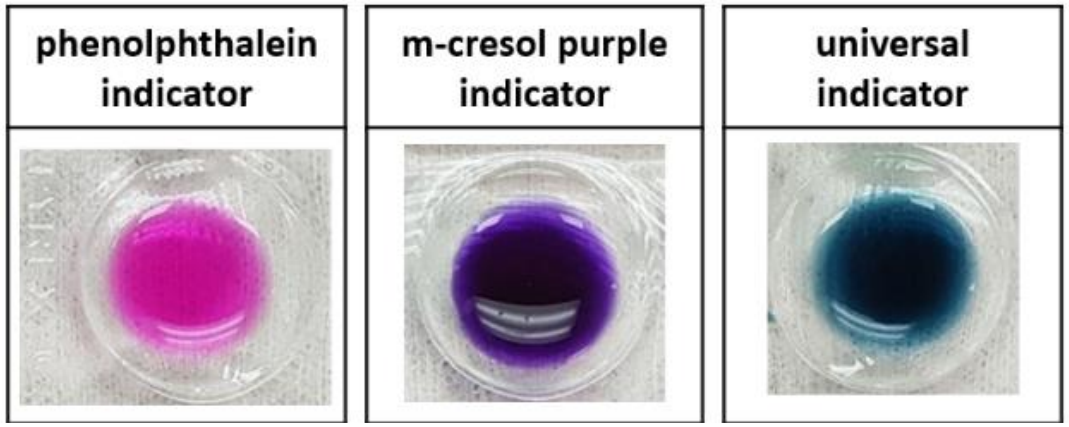
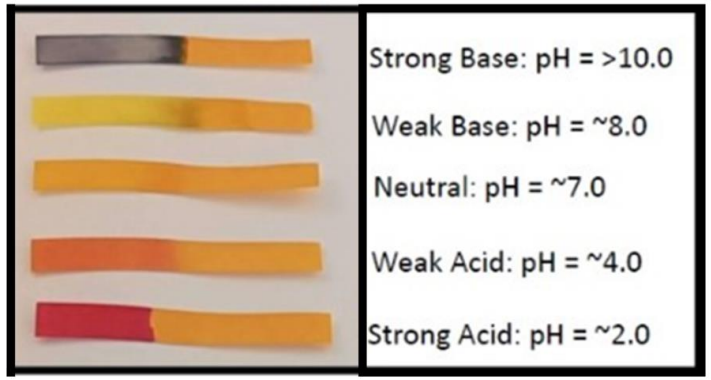
Alternative Existing Solutions



Incorporating Support While Maintaining Learning Outcomes

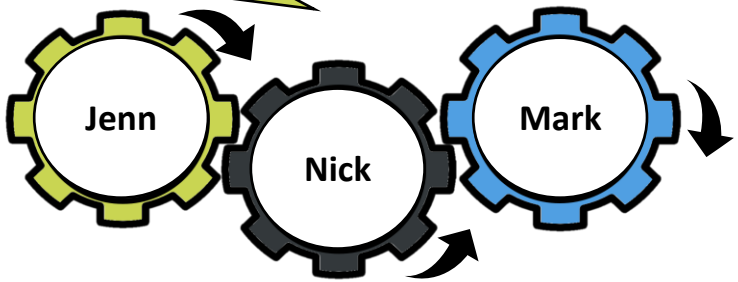


Universal pH Paper



"The online modules had helpful colour confirmations installed...but I noticed that they aren't available in the pre-lab assignment..." **- Student Fall 2021**

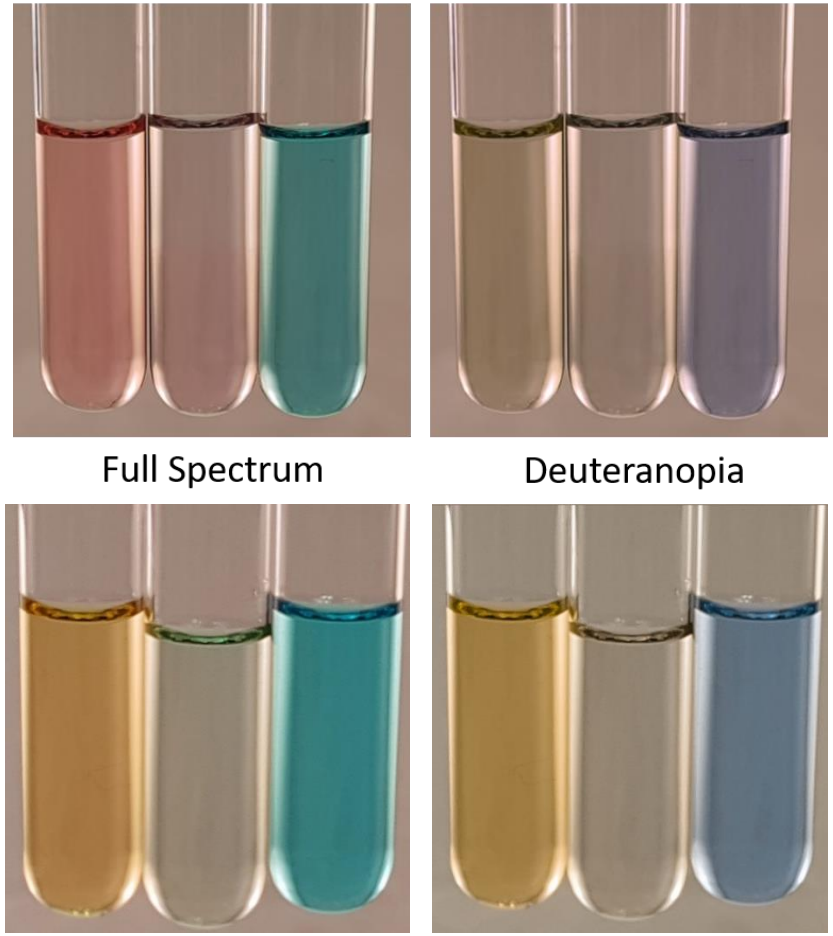
"That is a very good point about the pre-lab question! THANK YOU for this feedback. You have without a doubt helped many other students as well." **- Jenn**



"It works! I was able to complete it successfully! I am glad I was able to help current/future students who will encounter the same problem." **- Student Fall 2021**

Supporting Students and Educators (1)

Column Chromatography



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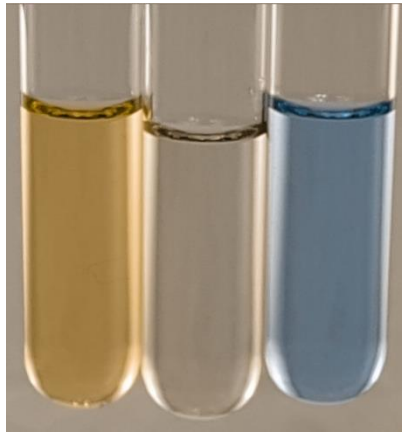
Deuteranopia



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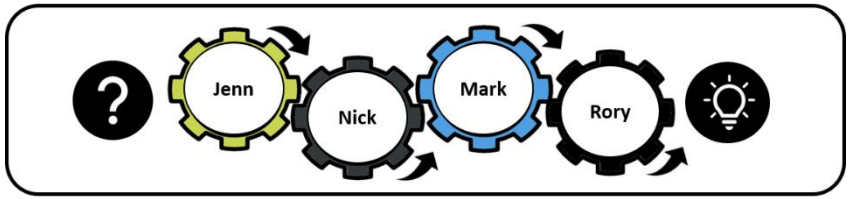
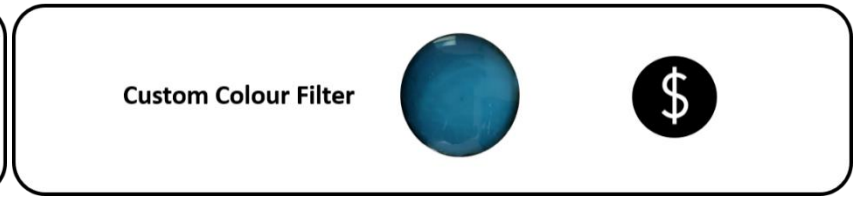
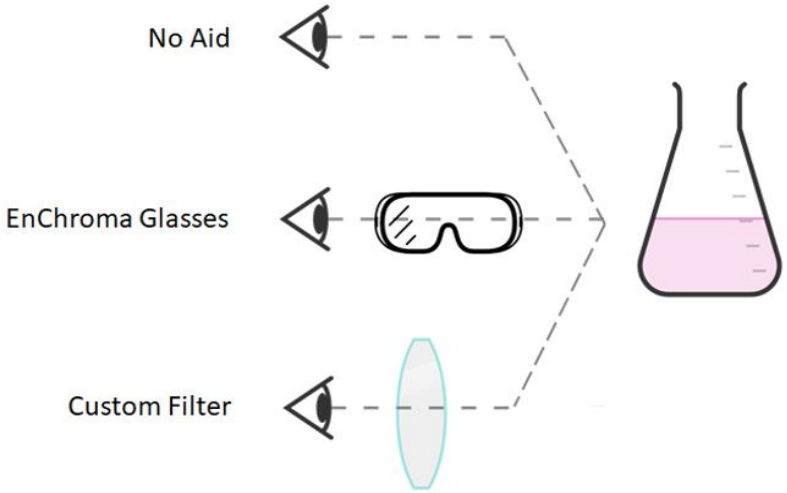
Red	Green	2-digit unknown number		
Yellow	Blue	2-digit unknown number	3-digit unknown number	Red/Green CVD (Protanopic/ Deuteranopic)
Yellow	Green	2-digit unknown number		

Explore (Or Create!) Low-Budget Solutions







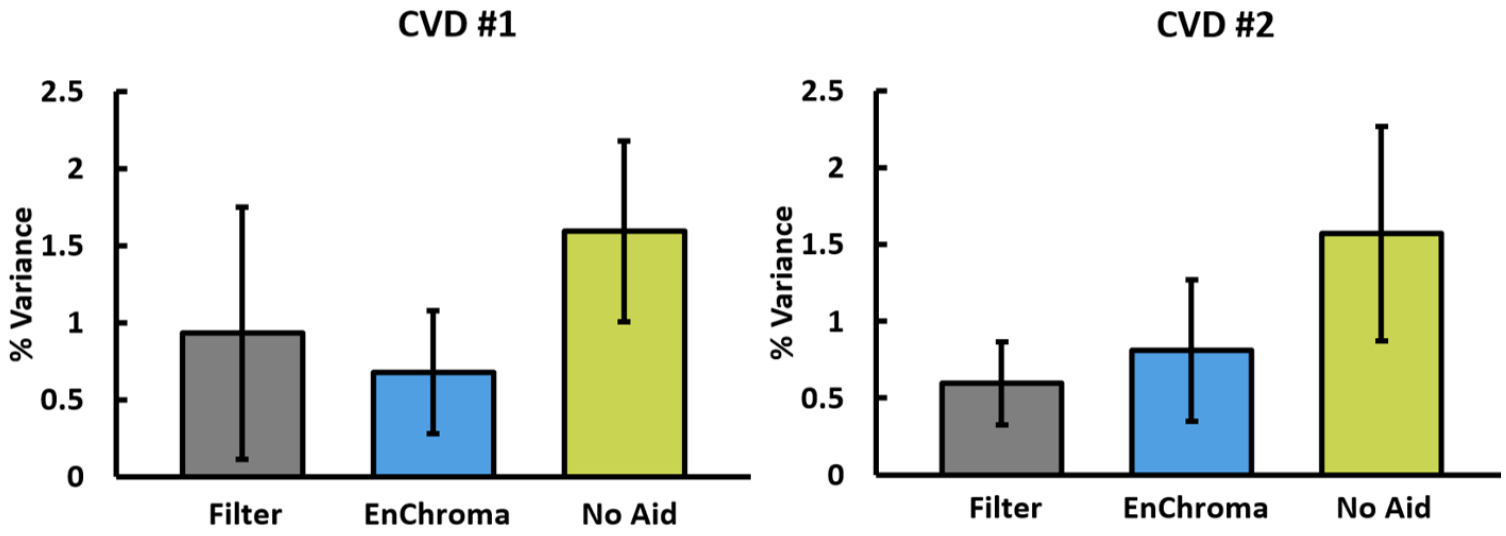


Figure 1: Comparison of % Variance in Moles of KHP From Actual Determined With Filter, EnChroma Glasses, and No Aid for CVD Individuals

(5) EnChroma

Starting Small and Dreaming BIG!

Nick

Katlyn

Jenn

Hey Jenn…what colour is this?

- Online/Blended Learning Support
- Teaching Team Support
- Rapidly evolving pilot work
- Community and Belonging
- …and maybe an OER!

PB PRESSBOOKS

Navigating Colour Vision Deficiency (CVD) in Science Education

Jennifer MacDonald/Nicholas Roberts/Katlyn Near

But Most Importantly, Community Within The Course

- Awareness, Access, Inclusion and Belonging

Exclusionary Language
"You probably can't help me..."
"I'm worried about..."
"I feel like I am cheating..."

Inclusionary Language
"Thank you for describing the colours...I'm colour-blind...how can I best navigate this lab?"
"[I felt] in control of my learning and experiments."
"Our team would like to opt for the colour-blind friendly unknown please."

"I really would like [my TA] to continue talking about his struggle with colour blindness and give us resources to help if we are having difficulty with said issue."

"How can I help or get involved with this project?!"

References and Resources

References

1. Colour Blind Awareness https://www.colourblindawareness.org/ (accessed Apr 20, 2022).
2. McIntyre, D. Colour Blindness: Causes and Effects; Dalton Publishing: Chester, 2002.
3. Chromatic Vision Simulator App: https://asada.website/cvsimulator/e/ (accessed Apr 20, 2022).
4. Coloring for Colorblindness: https://davidmathlogic.com/colorblind/ (accessed Apr 29, 2022).
5. Coblis Colour Blindness Simulator Coblis — Color Blindness Simulator – Colblindor (color-blindness.com) (accessed Apr 20, 2022).
6. WebAIM (Web Accessibility in Mind) Contrast Checker: https://webaim.org/resources/contrastchecker/
7. EnChroma Eyewear for Color Blindness https://enchroma.com/ (accessed Apr 20, 2022).
8. We Are Colourblind: https://wearecolorblind.com/ (accessed Apr 20, 2022).

Resources

tinyurl.com/ColoursChem

(link active until June 2, 2022)

An Open Educational Resource – In Construction

PB PRESSBOOKS

- Currently in the process of writing an opensource book!
- This is brand new, but keep your eyes peeled for something in about a year

Explore (Or Create!) Low-Budget Solutions

enchroma.

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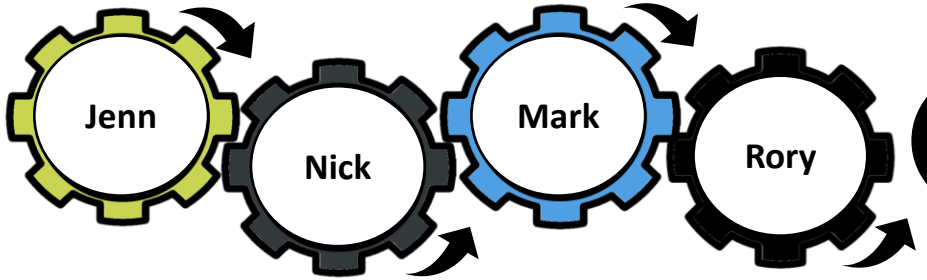
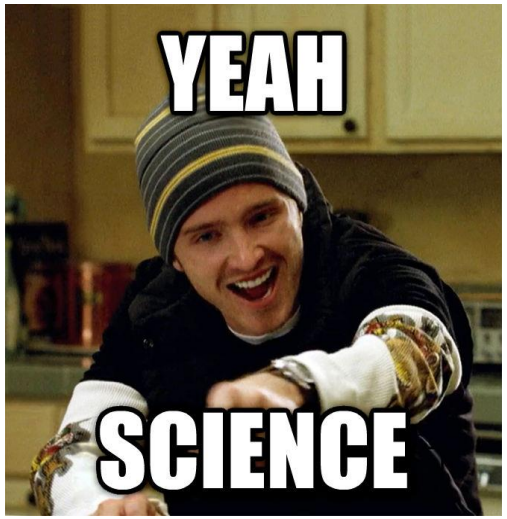


Custom Colour Filter

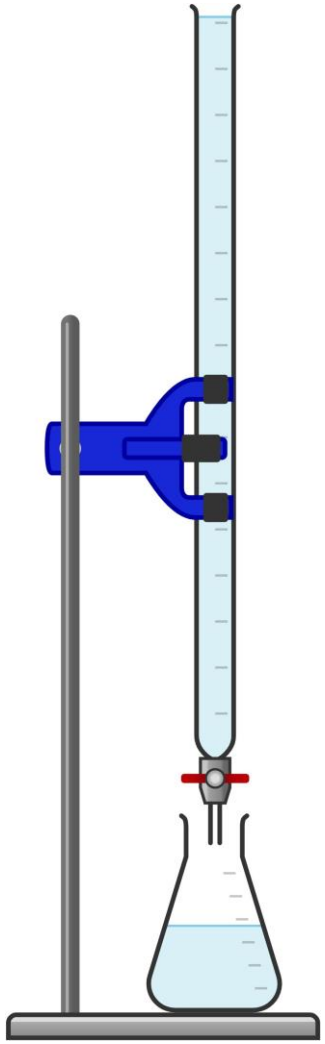
$

- Our preliminary data shows that EnChroma Glasses **AND** Custom Colour Filter produce an improvement in achieving experimental objective for CVD individuals.



(7) EnChroma

How Do We Quantitate Personal Colour Experience?

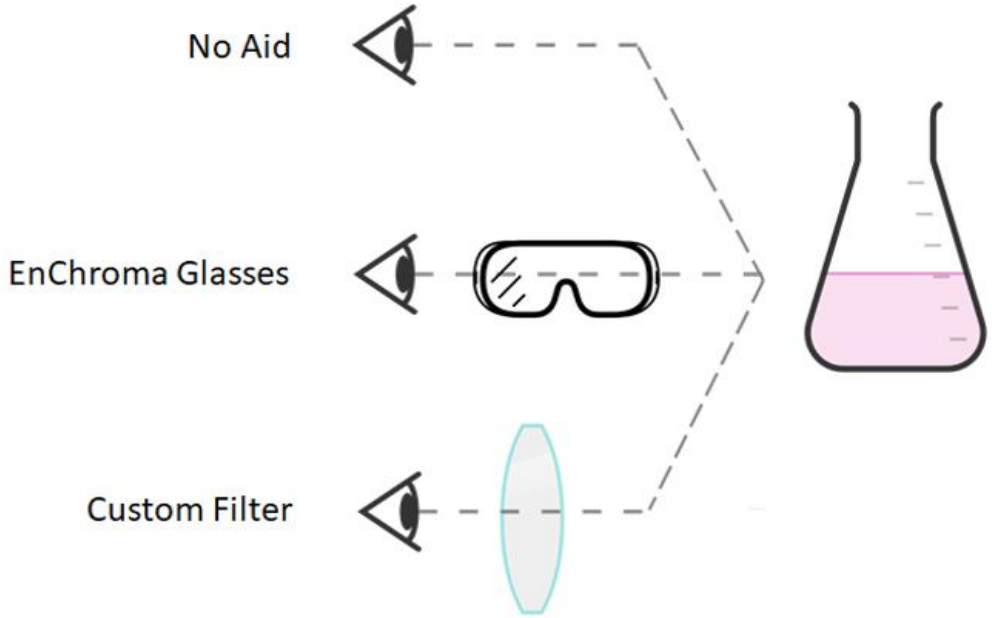


Potassium hydrogen phthalate (KHP) was titrated against standardized Sodium Hydroxide (NaOH)

~0.1 M NaOH
(standardized with pH probe)

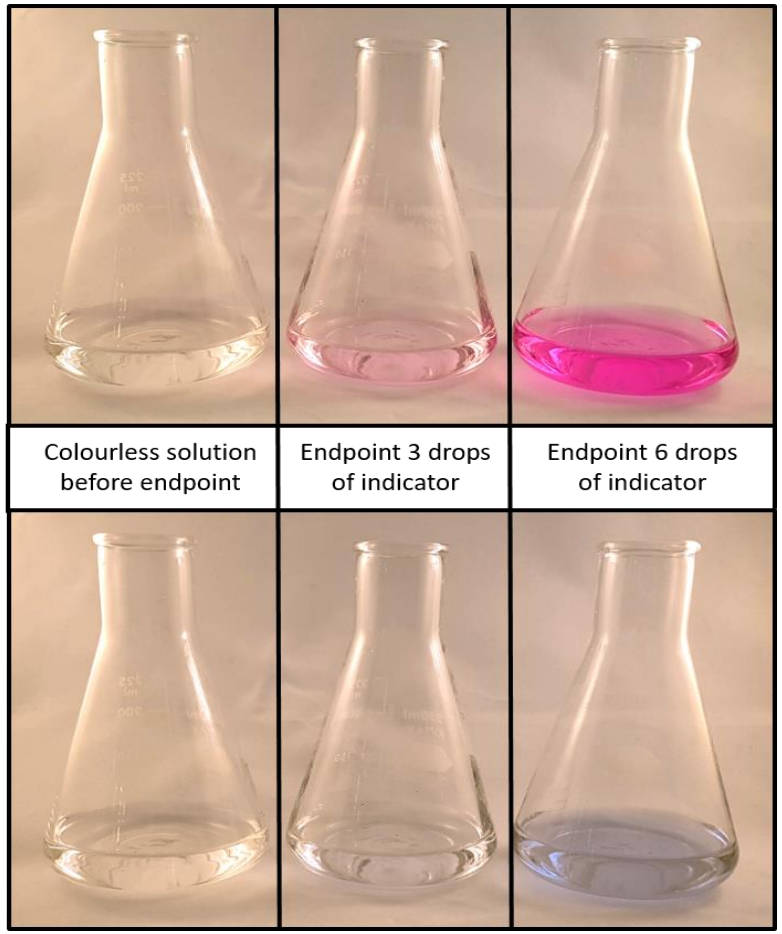
~0.5g KHP
3 drops phenolphthalein (0.1%)
25 mL distilled water

- Two groups of individuals:
  - CVD: n = 2
  - Control: n = 2
- Order of titrating was randomized
- Mass of KHP unknown during experiment



Colour Vision Deficiency Support

Full Spectrum

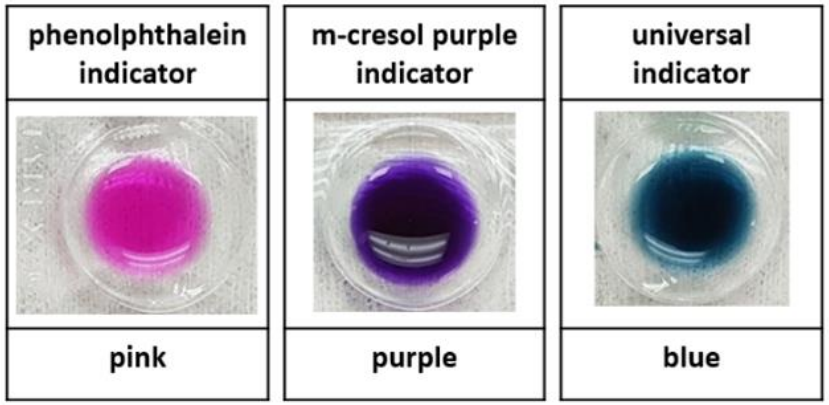


Deuteranopia

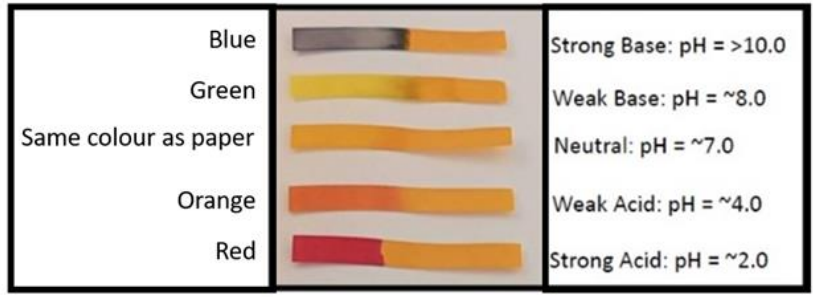
"**I feel like I am cheating** if I ask you to confirm that I have assigned my colours correctly."

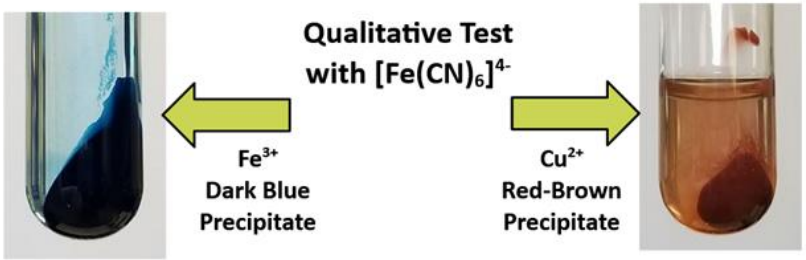
How can we help?

- Is the learning outcome how one sees colour, or how one interprets it?
- Provide modes of colour confirmation



Universal pH Paper

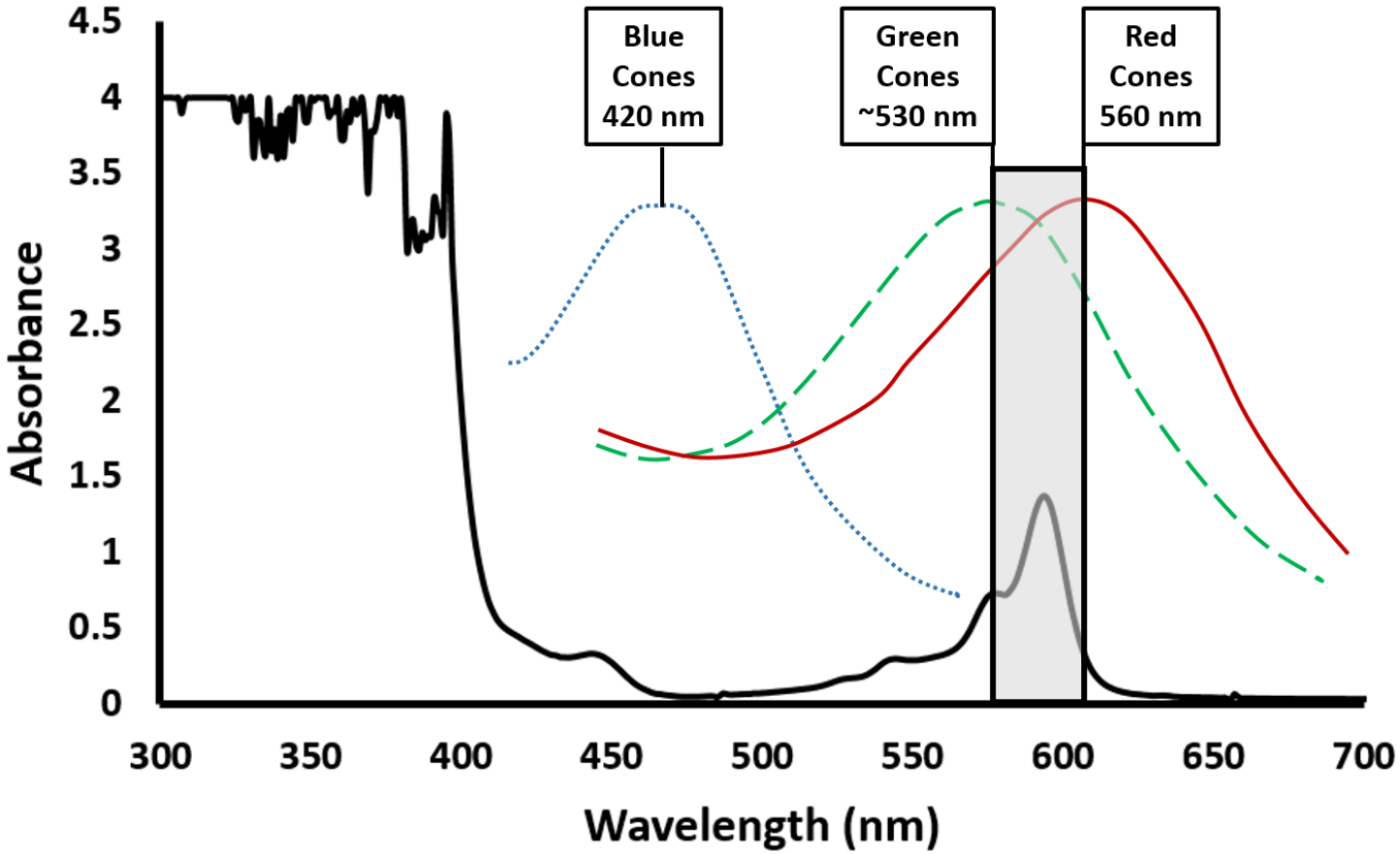




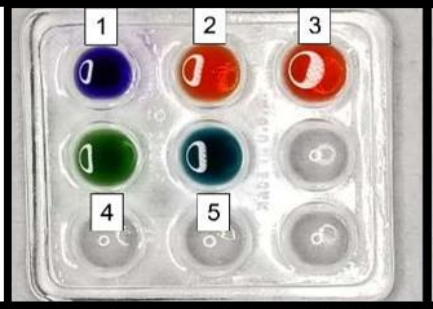
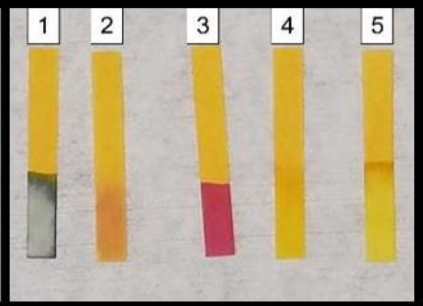
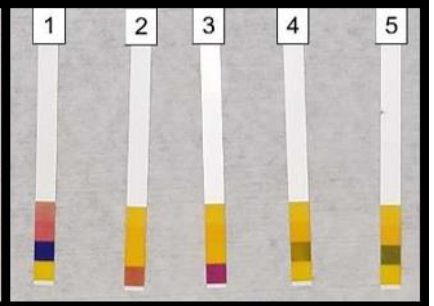
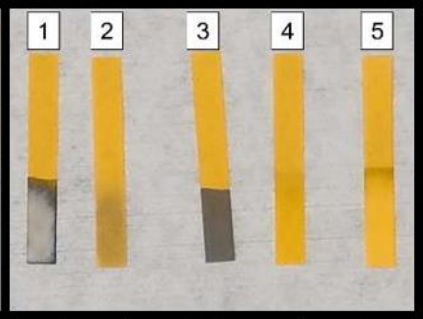
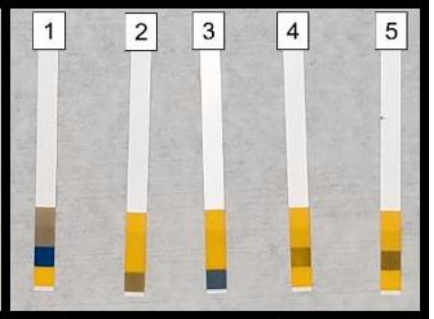
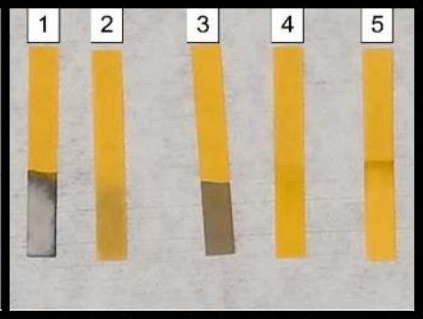
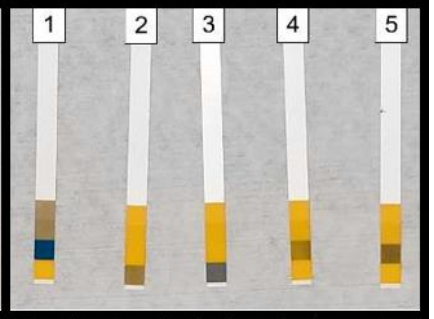
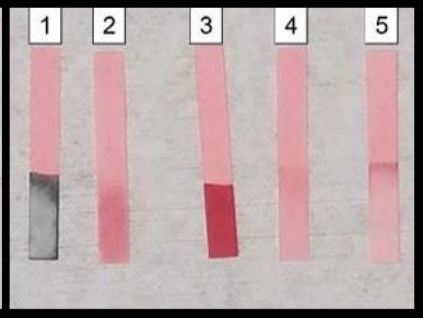
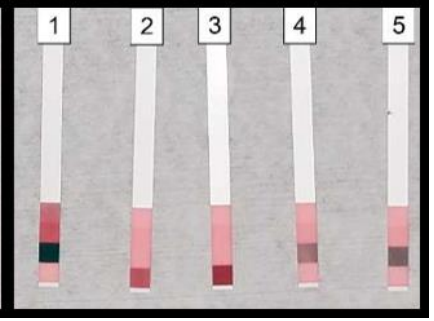
Commercially Available CVD Filter - EnChroma

enchroma

- Deuteranopia: missing green cones
- EnChroma Glasses “block” wavelengths



(2) McIntyre, (5) EnChroma, (6) We Are Colour Blind

	Universal pH indicator	Universal pH paper	Universal pH strips
Trichromatic			
Protanopic			
Deuteranopic			
Tritanopic			

		Unknown Numbers		
Dye 1	Dye 2	Main	CVD Support	CVD Type
Red	Blue	2-digit unknown number	3-digit unknown number	Blue/Yellow CVD (Tritanopic)
Red	Green	2-digit unknown number		
Yellow	Blue	2-digit unknown number	3-digit unknown number	Red/Green CVD (Protanopic/ Deuteranopic)
Yellow	Green	2-digit unknown number		

This is just Jenn making an image...leaving it here for now in case we find that we hate it and then I can fix it fast

This is just Jenn making an image...leaving it here in case we hate it and need to edit quick